Week-5

```
CREATE DATABASE PaymentDB;
USE PaymentDB;

-- Customer Table
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE
);

-- Orders Table
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    Order_Date DATE NOT NULL,
    Order_Amount DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID)
);

-- Payment Table
CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Payment_Date DATE NOT NULL,
    Payment_Mode VARCHAR(30) NOT NULL,
    Payment_Status VARCHAR(20) NOT NULL,
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    Transaction_Amount DECIMAL(10,2) NOT NULL,

    FOREIGN KEY (Order_ID)
        REFERENCES Orders(Order_ID),

    CHECK (Payment_Mode IN
    ('UPI', 'Credit Card', 'Debit Card',
     'Net Banking', 'Cash on Delivery')),

    CHECK (Payment_Status IN
    ('Pending', 'Success', 'Failed')),

    CHECK (Transaction_Amount >= 0)
);

-- Insert Customers
INSERT INTO Customer VALUES
(1, 'Arun', 'arun@gmail.com'),
(2, 'Priya', 'priya@gmail.com'),
(3, 'Kumar', 'kumar@gmail.com'),
(4, 'Divya', 'divya@gmail.com'),
(5, 'Ravi', 'ravi@gmail.com');

-- Insert Orders
INSERT INTO Orders VALUES
(101, 1, '2026-10-01', 1500),
(102, 2, '2026-10-02', 2000),
(103, 3, '2026-10-03', 1200),
(104, 4, '2026-10-04', 3000),
(105, 5, '2026-10-05', 2500);
```

```
-- Insert Payments
INSERT INTO Payment VALUES
(1, 101, '2026-10-01', 'UPI', 'Success', 1500),
(2, 102, '2026-10-02', 'Credit Card', 'Failed', 2000),
(3, 103, '2026-10-03', 'Debit Card', 'Pending', 1200),
(4, 104, '2026-10-04', 'Net Banking', 'Success', 3000),
(5, 105, '2026-10-05', 'Cash on Delivery', 'Pending', 2500);

-- 1. Display All Payments
SELECT * FROM Payment;

-- 2. Display Successful Payments
SELECT * FROM Payment
WHERE Payment_Status = 'Success';

-- 3. Display Failed and Pending Payments
SELECT * FROM Payment
WHERE Payment_Status IN ('Failed', 'Pending');

-- 4. Update Payment Status After Retry
UPDATE Payment
SET Payment_Status = 'Success'
WHERE Payment_ID = 2;

-- 5. Find Payment for Specific Order
SELECT * FROM Payment
WHERE Order_ID = 101;

-- 6. Count Successful and Failed Payments
SELECT Payment_Status, COUNT(*) AS Total
FROM Payment
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WHERE Payment_Status IN ('Success', 'Failed')
GROUP BY Payment_Status;

-- 7. Display Customer, Order and Payment Details
SELECT
   c.Customer_Name,
   o.Order_ID,
   o.Order_Date,
   p.Payment_ID,
   p.Payment_Mode,
   p.Payment_Status,
   p.Transaction_Amount
FROM Customer c
JOIN Orders o
ON c.Customer_ID = o.Customer_ID
JOIN Payment p
ON o.Order_ID = p.Order_ID;

-- 1. Number of Transactions by Payment Mode
SELECT Payment_Mode, COUNT(*) AS Total
FROM Payment
GROUP BY Payment_Mode;

-- 2. Most Preferred Payment Method
SELECT Payment_Mode, COUNT(*) AS Total
FROM Payment
GROUP BY Payment_Mode
ORDER BY Total DESC
LIMIT 1;

-- 3. Total Successful Payment Revenue
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SELECT SUM(Transaction_Amount) AS Total_Revenue
FROM Payment
WHERE Payment_Status = 'Success';

-- 4. Revenue by Payment Method
SELECT Payment_Mode,
    SUM(Transaction_Amount) AS Revenue
FROM Payment
WHERE Payment_Status = 'Success'
GROUP BY Payment_Mode;

-- 5. Average Transaction Amount
SELECT AVG(Transaction_Amount) AS Average_Amount
FROM Payment;

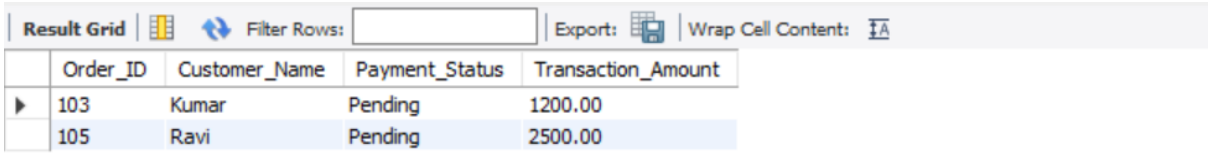
-- 6. Customer Payment History
SELECT c.Customer_Name,
    o.Order_ID,
    p.Payment_Date,
    p.Payment_Mode,
    p.Payment_Status,
    p.Transaction_Amount
FROM Customer c
JOIN Orders o
ON c.Customer_ID = o.Customer_ID
JOIN Payment p
ON o.Order_ID = p.Order_ID
ORDER BY c.Customer_ID, p.Payment_Date;

-- 7. Count All Payment Statuses
SELECT Payment_Status, COUNT(*) AS Total
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FROM Payment
GROUP BY Payment_Status;

-- 8. Orders with Failed or Pending Payments
SELECT o.Order_ID,
    c.Customer_Name,
    p.Payment_Status,
    p.Transaction_Amount
FROM Orders o
JOIN Customer c
ON o.Customer_ID = c.Customer_ID
JOIN Payment p
ON o.Order_ID = p.Order_ID
WHERE p.Payment_Status IN ('Failed', 'Pending');
```

Output:



Order_ID	Customer_Name	Payment_Status	Transaction_Amount
103	Kumar	Pending	1200.00
105	Ravi	Pending	2500.00